Mini Assessment - API Error Handling

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Q1. If an API returns 429, what should your program do?

A 429 status code means "Too Many Requests" - the client has hit the API's rate limit. The program should:
• Stop sending requests immediately and wait before retrying.
• Check the "Retry-After" header (if provided) and wait that long.
• Use a retry mechanism with exponential backoff (wait 1s, then 2s, then 4s, etc.).
• Avoid retrying endlessly - set a maximum number of retry attempts.

Q2. If response.json() fails, what should your program do?

response.json() fails when the response body is not valid JSON (e.g., an HTML error page or empty response). The program should:
• Catch the exception using a try-except block instead of letting the program crash.
• Log or print the raw response text to understand what was actually returned.
• Check the response status code and content-type before parsing.
• Show a proper error message to the user instead of failing silently.

Q3. If a JSON key is missing, why might .get() be safer than []?

Using [] on a missing key raises a KeyError and crashes the program. Using .get() is safer because:
• .get() returns None (or a default value you specify) instead of raising an error.
• It lets the program continue running smoothly even if the key doesn't exist.
• Example: data.get("name", "Unknown") returns "Unknown" if "name" is missing, instead of crashing.